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TECHNICAL SPECIFICATION

EARTH FAULT PASSAGE INDICATORS WITH SPLIT CORE CURRENT TRANSFORMER FOR 11 KV UNDERGROUND DISTRIBUTION SYSTEM

Specification No.	KEJ01308:2012	Revision No.	2
Prepared by	Rahimah Uzir Engineer (Protection) Asset Management Department TNB, Distribution Division		
Approved by	TNB, Distribution Division Technical Committee Dated: 30th November 2012		

(WAN NAZMY BIN WAN MAHMOOD)
General Manager (Engineering Services)
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1.0 Scope

This specification covers the requirements of Earth Fault Passage Indicators for use on 11 kV, 50 Hz, 3-phase, 3-wire underground distribution systems.

The Earth Fault Passage Indicators shall be designed for use in conjunction with automatic or non-automatic oil or gas insulated ring main units or circuit breakers controlling cable networks where the networks operate as a radial or open-ring distribution system without interconnection between substations on the medium voltage side.

This earth fault passage indicator, where installed, will provide a means of indicating the passage of fault current along the cable systems and the section of line where the fault has occurred enabling quick location, sectionalisation, isolation of the faulted section and restoration of supply to healthy circuits.

2.0 System Conditions

The equipment shall be suitable for use on distribution systems in which the neutral point is earthed directly or by means of a resistor and in which the maximum short time current corresponds to rated short-circuit levels of maximum 25kA at 11 kV, the rated short-circuit duration being 3 seconds. The equipment shall perform satisfactorily without suffering damage under these conditions.

A maximum variation in system frequency of 2 Hz from the nominal frequency of 50 Hz shall not result in variations in the operation or resetting characteristics of the fault passage indicator.

3.0 Climatic Conditions

The equipment shall be suitable for use in tropical weather conditions with high humidity of 85% to 95%, heavy rainfall and ambient temperatures of 28°C to 45°C.

4.0 Voltage Rating Of Equipment

Equipment intended for direct connection or indirect coupling e.g. by inductive or capacitive coupling to 11 kV systems shall be capable of functioning satisfactorily and without suffering from damage for the full range of steady state and transient voltages likely to be encountered in service of such systems. Such equipment shall be capable of withstanding, without damage, the application of test voltage specified in IEC 60502-2 for 11 kV cables.

The secondary winding and associated connections for the fault passage indicator shall be capable of withstanding the insulation voltage tests specified in IEC 60044-1(Part 1, clause 5.1.4).

5.0 Components Of The Earth Fault Passage Indicator

A complete unit of earth passage indicator shall consist of.

a. A split-core current transformer
b. Detection unit (include with internal signaling unit)
c. External Signaling unit – Optional

6.0 Current Transformer

The current transformer for the detection unit shall be an encapsulated split-core design suitable for embracing the sheath of:

Table 1.0: Current Transformer (CT) Size

No.	Cable Type	Diameter of CT
1	3 core cable (conductor section up to 300sq.mm)	Maximum diameter – 90mm
2	3 single core cable (conductor section up to 500sq. mm)	Maximum diameter – 300mm

Please Refer Addendum for CT Diameter

The current transformer shall also suitable for use on outdoor installation.

The current transformer shall be designed to saturate during heavy earth faults, to prevent and protect possible damage to the indicating elements at the higher current levels.

The current transformer encapsulation shall be made of a durable, chemically inert, UV and weatherproof polymeric material that will provide the necessary mechanical robustness. Straps, if used, shall be made of stainless steel.

For toroidal type current transformer method of securing the current transformer to the cable may be of straps made of UV protected material.

The current transformer shall be able to withstand periodic immersions in water over medium duration periods (1 to 2 weeks) as in flood prone areas, without unduly affecting performance.

7.0 Detection Unit

The fault passage indicator is required to detect earth fault currents down to a value of at least 40 amperes.

The detection relay shall provide for multiple, discrete user settable earth fault current pick-up values with a minimum range at least from 40A to 240A. It shall provide for user settable operation delay time with a minimum range from 50mS to 150mS.

The detection unit shall be mountable on the inside or outside wall of an indoor type substation or on a compact substation. The enclosure shall be at least of protection index IP54 to IEC 529.

The detection unit shall be powered by;

a. Battery
b. External DC (up to 30V DC) - optional
c. 230Vac auxiliary LV supply available from the substation– optional

8.0 Signaling Device

The indicator may be a separate unit in itself or form an integral part of the whole device.

Indication of the passage of fault current shall be by means of a light indicator which operates and remains in operated position until reset. The indicator may be mounted on the outside wall of an indoor type 11kV distribution substation or on some support for an outdoor substation remote from the detection relay. It shall be suitable for fixing with screws or stainless steel band straps.

The illumination of the light shall be such that it is clearly visible from a minimum distance of 8 metres in broad daylight and in tropical conditions e.g. thunderstorm.

The light indicator shall NOT be of the filament type to eliminate the necessity of maintenance and replacement. It shall preferably be of the high intensity light emitting diode (red).

The light indicator shall be powered by a minimum of one internal long life battery of lithium type or equivalent with minimum 2.5Ah with at least 10 years of operation life span and easily replaceable. The supplier shall provide complete details of the battery and battery lifespan calculation inclusive of permanent contact of SCADA duty (Eg: 12 fault per years, with 4 hours earth fault indicator operation). The calculation shall not used external power for the source of supply.

The light indicator shall be fully weather proof, resistant to corrosive atmosphere, being designed for outdoor installation and meeting levels of protection IP54 to IEC 529.

An effective means of separation shall be provided between the L.V a.c supply - and the detection circuit to ensure that the circuit board is protected against any possible rise in voltages or voltage surges of 6 kV.

Voltage free contacts reflecting the status of the earth fault passage indicators shall be provided for SCADA applications. The latching period for the contacts must be permanent contact.

9.0 Resetting

The activated fault passage indicators shall automatically reset on restoration of supply. The resetting time shall ensure that the affected section has been energised for a period of not less than 10 seconds before automatic reset takes place. The total resetting time shall not exceed 30 seconds.

The detection relay and its associated signaling device shall be able to:

a. Manually reset to its original state

b. Dual in-line package (DIP) switch setting (eg: 4hrs, 8hrs, 16hrs or 32 hrs)

c. Supervisory reset from the SCADA. The requirement are as below:

- SCADA will provide input contact for 2 seconds pulse in the range of 9 to 38Vdc input to the EFI contacts for the purpose of supervisory EFI reset command. For SCADA reset EFI shall immediately reset after received commands from SCADA.

10.0 Cables

Please Refer Addendum for updates in Clause 10

The split-core current transformer shall be suitable for installation onto the 11kV underground cable network consisting of the following types and sizes of cable.

a. *Single Core Cable*

i. XLPE – maximum size of cable is up to 500sqmm (with reference to clause 6(b))

b. *3 Core Cable*

ii. PILCDSTA – maximum size of cable is up to 300 sq. mm.

iii. XLPE – maximum size of cable is up to 240 sq. mm

The supplier shall have to be well versed with the construction of the cable to ensure that the proper connection of the earth wire for both types of cables can be clearly illustrated in the installation instructions.

11.0 Battery Check Indicator

Battery check indicator in a form of push button shall be included as part of the EFI. This is as an additional feature to assist on EFI maintenance. The battery check signal shall consist of Light Emitting Diode (LED – amber).

12.0 Type Test

The tests are included with the protection degree of enclosure of the product and the environmental test of the product. The product shall have passed the type test as follows:

a. IP verifications as per IEC 60529

The type test as mentioned above are to be conducted at any national or international independent laboratory accredited to ISO/IEC 17025 by the National Accreditation Body who is the MRA/MLA signatory to ILAC, APLAC, EA, NACLA or IAAC. The test to be conducted shall be within the scope of accreditation of the laboratory and shall cover the appropriate capacity limit for the specified test.

TNB shall reserve the rights to witness the conduct of the said test if deemed necessary through its engineers or appointed representative.

Note:

a. MRA : Mutual Recognition Agreement
b. MLA : Multilateral Agreement
c. ILAC : International Laboratory Accreditation Cooperation
d. APLAC : Asia Pacific Laboratory Accreditation Cooperation
e. EA : European Cooperation for Accreditation
f. NACLA : National Cooperation for Laboratory Accreditation
g. IAAC : Inter-American Accreditation Cooperation

13.0 Routine Test

Test shall be carried out by the manufacturer on all product in the finished state or, where appropriate, during the manufacturing processes. The tests that shall be made on the EFI are:

a. Physical and dimensional check on dimension and construction
b. The EFI cover can be easily opened
c. The specified battery is installed in the unit
d. The battery can be easily installed.

14.0 Sampling Test

The manufacturer shall carry out sampling test on samples of product in the finished state. The sampling tests that shall be made on the EFI are:

a. Physical check on the dimension and marking
b. DIP switch bit position for default setting as agreed by TNB.
c. Current injection test to prove current trip setting value

Sampling rate shall be as required in the tender requirement.

15.0 Installation Instructions

The supplier shall provide clearly illustrated detailed installation procedures with each complete set of the Earth Passage Indicators.

The installation procedure shall be accordance to TNB cabling installation and methods of termination and earthing of cables in the TNB 11 kV cable network so that the correct installation procedures or alternatives can be provided.

The clearly illustrated detailed installation procedure shall be provided for each unit of Earth Fault Indicator.

The supplier shall also, provide the setting sticker inside of the Detection Unit, to facilitate the EFI setting record.

End of Document

SALINAN TNB

TNB EVALUATION CRITERIA

INDOOR TYPE EARTH FAULT PASSAGE INDICATORS WITH SPLIT CORE CURRENT TRANSFORMER FOR 11 kV UNDERGROUND DISTRIBUTION SYSTEM

1. Mandatory Requirement

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
a. Earth Fault Indicator Mandatory Requirements declaration (must be supported and highlighted in submitted documentation, i.e. from technical brochure, manual, etc):	i. Humidity Operating Range : **85% - 95%**	
	ii. Temperature Operating Range: **28°c - 45°c**	
	iii. Sensing Method (Direct/Indirect): **Direct**	
	iv. Control and Signalling Unit: **Separate/Combine**	
	v. State Auxiliary Power for Control Unit (230 V): **230V**	
	vi. Min Primary E/F Current Pick-Up Value: **40A**	
	vii. Max Primary E/F Current Pick-Up Value: **240A**	
	viii. Min Time Delay: **Range from 50mS to 150mS**	
	ix. Battery Sizing (Ah): **Minimum 2.5Ah**	
	x. Battery details & battery lifespan calculation provided (12 fault per yrs with 4 hrs EFI operation): **Y**	
	xi. Type of Light Indication (Bulb,LED-Red,Etc): **LED-Red**	
	xii. Latching Period for the EFI contact (Permanent/Unlatch): **Permanent**	
	xiii. Distance of Visiblity of Indication: **8 metres**	

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
	xiv. Color of Light Indication: **Red**	
	xv. Battery Check Indicator: **Amber**	
	xvi. Power Source for Indication: **Battery**	
	xvii. Method of Resetting Indication: **Manual, Auto, Time (DIP Switch) & Supervisory Reset from SCADA**	
	xviii. State Auxiliary Power for Control Unit (resetting of EFI): **230Vac**	
	xix. Range of DIP Switch Setting for EFI resetting: **4hrs, 8hrs, 16hrs, 32hrs**	
	xx. Voltage Free Contact for Scada (Y/N): **Y**	
	xxi. Time Delay for supervisory reset from SCADA: **immediate/2s**	
	xxii. Voltage Range of EFI terminal can accept for supervisory reset from SCADA: **9 – 38Vdc**	
	xxiii. Installation Manual Provided: **Y**	
b. Current transformer of EFI Requirements declaration (must be supported and highlighted in submitted documentation, i.e. from technical brochure, manual, etc):	i. Current Transformer Type: **Core Balance**	
	ii. Split Current Transformer(Y/N): **Y**	
	iii. Max CT External Diameter: **3core core(>90 sq. mmp), 3 single core(500 sq. mmp)**	Please Refer Addendum Table for Declaration
	iv. CT can withstand periodic immersion in water (1 to 2 weeks): **Y**	
	v. Method of Securing CT	

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
	vi. CT Encapsulated	
c. Type test report/certificate submitted. Report /certificate submitted shall be from accredited laboratory or in house laboratories witness by a certified third party.	i. Report/Certificate from accredited laboratory: **Y**	
	ii. Degree of protection of Control unit (IP Rating)– compliance to IEC 529: **IP 54**	
	iii. Degree of protection of Signalling Unit (IP Rating)– compliance to IEC 529: **IP 54**	
	iv. IP rating declaration: **IP 54 to IEC 529**	
d. Cable Length:	i. Cable length from CT to Controller Unit: **Min. 16 meter**	

2. Additional Info

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
a. Earth Fault Indicator additional Info declaration (must be supported and highlighted in submitted documentation, i.e. from technical brochure, manual, etc):	i. Dimension of Control Unit	
	ii. Dimension of Signalling Unit	
	iii. Step of Selectivity Of The E/F Setting	
	iv. Step of Selection Of Time Delay	
	v. Max & Min Time for Indication	
	vi. Selectable range of Time delay	
	vii. Current Transformer Ratio	
	viii. Manufacturer	
	ix. Country of Origin	
	x. Type of Model No/Trade Mark	

SALINAN VENDOR

TNB EVALUATION CRITERIA

INDOOR TYPE EARTH FAULT PASSAGE INDICATORS WITH SPLIT CORE CURRENT TRANSFORMER FOR 11 kV UNDERGROUND DISTRIBUTION SYSTEM

1. Mandatory Requirement

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
a. Earth Fault Indicator Mandatory Requirements declaration (must be supported and highlighted in submitted documentation, i.e. from technical brochure, manual, etc):	i. Humidity Operating Range :	
	ii. Temperature Operating Range:	
	iii. Sensing Method (Direct/Indirect):	
	iv. Control and Signalling Unit (Separate/Combine):	
	v. Min Primary E/F Current Pick-Up Value:	
	vi. Max Primary E/F Current Pick-Up Value:	
	vii. Min Time Delay:	
	viii. Battery Sizing (Ah):	
	ix. Battery details & battery lifespan calculation provided (12 fault per yrs with 4 hrs EFI operation):	
	x. Type of Light Indication (Bulb,LED-Red,Etc):	
	xi. Latching Period for the EFI contact (Permanent/Unlatch):	
	xii. Distance of Visibility of Indication:	
	xiii. Colour of Light Indication:	
	xiv. Battery Check Indicator Provided (Y/N):	
	xv. Power Source for Indication:	

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
	xvi. Method of Resetting Indication:	
	xvii. State Auxiliary Power for Control Unit (resetting of EFI):	
	xviii. Range of DIP Switch Setting for EFI resetting:	
	xix. Voltage Free Contact for SCADA (Y/N):	
	xx. Time Delay for supervisory reset from SCADA:	
	xxi. Voltage Range of EFI terminal can accept for supervisory reset from SCADA:	
	xxii. Installation Manual Provided (Y/N):	
b. Current transformer of EFI Requirements declaration (must be supported and highlighted in submitted documentation, i.e. from technical brochure, manual, etc):	i. Current Transformer Type:	
	ii. Split Current Transformer(Y/N):	
	iii. Max CT External Diameter:	
	iv. CT can withstand periodic immersion in water (1 to 2 weeks):	
	v. Method of Securing CT	
	vi. CT Encapsulated	
c. Type test report/certificate submitted. Report /certificate submitted shall be from accredited laboratory or in house laboratories witness by a certified third party.	i. Report/Certificate from accredited laboratory (Y/N):	
	ii. Degree of protection of Control unit (IP Rating)– compliance to IEC 529:	
	iii. Degree of protection of Signalling Unit (IP Rating)– compliance to IEC 529:	

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
	iv. IP rating declaration:	
d. Cable Length:	i. Cable length from CT to Controller Unit: **Min. 16 meter**	

2. Additional Info

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
a. Earth Fault Indicator additional Info declaration (must be supported and highlighted in submitted documentation, i.e. from technical brochure, manual, etc):	i. Dimension of Control Unit	
	ii. Dimension of Signalling Unit	
	iii. Step of Selectivity Of The E/F Setting	
	iv. Step of Selection Of Time Delay	
	v. Max & Min Time for Indication	
	vi. Selectable range of Time delay	
	vii. Current Transformer Ratio	
	viii. Manufacturer	
	ix. Country of Origin	
	x. Type of Model No/Trade Mark	

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ADDENDUM OF TECHNICAL SPECIFICATION

EARTH FAULT PASSAGE INDICATORS WITH SPLIT CORE CURRENT TRANSFORMER FOR 11 KV UNDERGROUND DISTRIBUTION SYSTEM

Specification No.	KEJ01308:2012	Revision No.	2
Prepared by	Idris Bin Yussof Senior Engineer (Protection System Development) Asset Management Strategy & Design Standard TNB, Distribution Network Division		
Approved by	TNB, Distribution Network Division Technical Committee Dated: 28 June 2019		

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Addendum of Technical Specification Earth Fault Passage Indicators With Split Core Current Transformer For 11 kV Underground Distribution System			TENAGA NASIONAL
Specification No.	KEJ01308:2012	Revision No. 2	Page 1 of 2
Approved by	TNB, Distribution Network Division Technical Committee Dated 28 June 2019		

The addendum are as follows :

<table>
<tr><th>No</th><th>Clause in existing Spesification</th><th>Addendum</th></tr>
<tr><td>1.</td><td>6.0
Current
Transformer</td><td>The current transformer for the detection unit shall be an encapsulated split-core design suitable for embracing the sheath of TNB cables as follows:

<table>
<tr><th>No.</th><th>Cable Type</th><th>Diameter of CT</th></tr>
<tr><td>1</td><td>3 core cable (conductor size up to 500sq.mm)</td><td>Diameter range of 110mm – 140mm</td></tr>
<tr><td>2</td><td>3 single core cable (conductor size up to 500sq. mm)</td><td>Diameter range of 290mm – 310mm</td></tr>
</table>

The EFI's current transformer shall meet the criteria in the table above and also shall be able to be installed (clamped) easily and properly before being considered as fit-for-use. The diameter of CT shall consider reasonable clearence for clamping procedure during installation process.

**continue with existing spec</td></tr>
<tr><td>2.</td><td>10.0
Cables</td><td>The split-core current transformer shall be suitable for installation onto the 11kV underground cable network consisting of the following types and sizes of cable (but not limited to):
a. Single Core Cable
i. XLPE – maximum size of cable is up to 500sqmm (with reference to clause 6.0)
b. 3 Core Cable
ii. PILCDSTA – maximum size of cable is up to 300 sq. mm.
iii. XLPE – maximum size of cable is up to 500 sq. mm

The supplier shall have to be well versed with the construction of the cable to ensure that the proper connection of the earth wire for both types of cables can be clearly illustrated in the installation instructions. With the understanding of the cable construction, the CT diameter shall be suitable for use onto the cables in the TNB system and with any kind of switchgear design.</td></tr>
</table>

***BOLD font represent the ammendment or additional items from existing specification**

ADDENDDUM FOR TNB EVALUATION CRITERIA
INDOOR TYPE EARTH FAULT PASSAGE INDICATORS WITH SPLIT CORE CURRENT TRANSFORMER FOR 11 kV UNDERGROUND DISTRIBUTION SYSTEM

1. Mandatory Requirement

CRITERIA	TNB REQUIREMENT	TENDERER RESPOND
b. Current transformer of EFI Requirements declaration (must be supported and highlighted in submitted documentation, i.e. from technical brochure, manual, etc):	*iii. For 3 Cores (up to 500sq.mm) Cables ;* *CT Diameter:* *110mm – 140mm*	
	iv. For 3-Single Core (up to 500sq.mm) Cables; *CT Diameter:* *290mm- 310mm*	